
師大微積分乙 114-102 期末考歷屆

► 積分技巧與基本計算

1. (師大微乙 (一)114#3) Evaluate the integrals. (計算積分)

(a) $\int_0^{\pi/2} \cos^3(3\theta) \sin^2(6\theta) d\theta$

(b) $\int x^3 \sin(2x) dx$

(c) $\int \frac{2x^2 - 6x + 4}{(x^2 + 1)(x - 1)^3} dx$

(d) $\int \tan^{-1}(\sqrt{x}) dx$

(e) $\int \sqrt[3]{\tan x} dx$

(f) $\int \ln(x) \cdot x^{\ln(x)-1} dx$

(g) $\int_0^{2025} \frac{\sqrt{x}}{\sqrt{x} + \sqrt{2025 - x}} dx$

(h) $\int \frac{\sin^2(x)}{1 + \cos^2(x)} dx$

2. (師大微乙 (一)113 暑 #2) Evaluate the following integrals.

(a) $\int \left(x + \sqrt{x} + 1 + \frac{1}{\sqrt{x}} + \frac{1}{x} \right) dx$

(b) $\int (x^2 \sin x + \cos^2 x) dx$

(c) $\int (\tan^2 x + \tan^4 x) dx$

(d) $\int \frac{2x}{\sqrt{1+x^4}} dx$

(e) $\int \frac{e^x}{e^{2x} - 3e^x + 2} dx$

(f) $\int_{-1}^1 (x^3 + 3)\sqrt{4 - x^2} dx$

3. (師大微乙 (一)113 暑 #3) Let a be a real number and $a > 0$.

(a) Evaluate $\int x \ln x dx$

(b) If a satisfies the equation $\int_0^a x \ln x dx = 0$, find the value of a .

4. (師大微乙 (一)113#2) Evaluate the integrals.

- (a) $\int \frac{\cos^5 x}{\sin^2 x} \, dx$
- (b) $\int \frac{x^2}{\sqrt{49 - x^2}} \, dx$
- (c) $\int \frac{2x^3 + 4}{(x - 1)^2(x^2 + 1)} \, dx$
- (d) $\int \frac{\tan x(\sec^2 x + 4 \sec x)}{\sec^2 x + 5 \sec x - 6} \, dx$
- (e) $\int_{-1}^1 \sqrt{x^{10}} + x^2 \sec x \tan x + \frac{1}{1 + x^2} \, dx$

5. (師大微乙 (一)113#6)

- (a) Evaluate $\int_0^1 \ln x \, dx$. (Note: this integral is an improper integral(瑕積分).)
- (b) Let a be a real number and $a > 0$. If a satisfies the equation $\int_0^a \ln x \, dx = 0$, find a .

6. (師大微乙 (一)112#2) Evaluate the following integrals.

- (a) $\int \frac{x^2}{x^4 - 1} \, dx$
- (b) $\int_0^{\pi^2/4} \cos \sqrt{x} \, dx$
- (c) $\int_0^e \ln x \, dx$
- (d) $\int \sec^5 x \, dx$
- (e) $\int \frac{(1 - y^2)^{5/2}}{y^8} \, dy$

7. (師大微乙 (一)111#1) Evaluate the integrals.

- (a) $\int \frac{\ln(\ln x)}{x \ln x} \, dx$
- (b) $\int_0^1 x \ln x \, dx$
- (c) $\int \frac{x^3}{x^2 + 2x + 1} \, dx$
- (d) $\int \frac{x}{\sqrt{1 + x^4}} \, dx$
- (e) $\int \sqrt{\sin x} \cos^3 x \, dx$
- (f) $\int \frac{x e^{2x}}{(2x + 1)^2} \, dx$

$$(g) \int \frac{\sec^2 \theta}{\tan^3 \theta - \tan^2 \theta} d\theta$$

$$(h) \int_{-\infty}^1 \frac{9r^2}{\sqrt{1-r^3}} dr$$

8. (師大微乙 (一)110#1) 計算下列積分. (Evaluate the following integrals.)

$$(a) \int_1^4 \left(\frac{2}{x^3} - \sqrt{x} \right) dx$$

$$(b) \int x^2 \sqrt{x^3 + 1} dx$$

$$(c) \int_0^3 \frac{x}{\sqrt{x+1}} dx$$

$$(d) \int \cos \sqrt{x} dx$$

$$(e) \int e^x \cos 2x dx$$

$$(f) \int \frac{2}{16x^2 - 1} dx$$

$$(g) \int \sqrt{9 - x^2} dx$$

$$(h) \int_4^6 \frac{1}{\sqrt{8x - x^2}} dx$$

9. (師大微乙 (一)110#5) 計算以下瑕積分. (Evaluate the following improper integrals.)

$$(a) \int_0^{\infty} x e^{-2x} dx$$

$$(b) \int_0^1 \ln x dx$$

10. (師大微乙 (一)109#1) Evaluate the integrals.

$$(a) \int \sqrt{16 - 4x^2} dx$$

$$(b) \int_1^{\infty} \frac{1}{x(\ln x)^3} dx$$

$$(c) \int x^4 \cos(2x) dx$$

$$(d) \int \sin(7x) \cos(3x) dx$$

$$(e) \int \frac{\sqrt{x^2 + 3}}{x} dx$$

$$(f) \int \sec(2x) \tan^2(2x) dx$$

(g) $\int_0^{\pi/2} \frac{\cos(\theta)}{(\sin^2(\theta) + 4\cos^2(\theta))^{3/2}} d\theta$

(h) $\int \frac{5x^2 + 20x + 6}{x^3 + 2x^2 + x} dx$

11. (師大微乙 (一)108#3)

(a) Does the improper integral $\int_1^\infty \frac{\ln t}{2\sqrt{t}} dt$ converge or diverge? Give reasons for your answer.

(b) Use the part (a) to evaluate $\lim_{x \rightarrow \infty} \frac{\int_1^x \frac{\ln t}{2\sqrt{t}} dt}{\sqrt{x^3}}$.

12. (師大微乙 (一)108#4) Find each integral.

(a) $\int \frac{8x}{x^3 + x^2 - x - 1} dx$

(b) $\int \tan^4(5x) dx$

(c) $\int e^x \sqrt{1 - e^{2x}} dx$

(d) $\int e^x \cos x dx$

(e) $\int_0^{\pi/2} \frac{\cos^5 x}{\sin^5 x + \cos^5 x} dx$ (Hint: Use the substitution $u = \frac{\pi}{2} - x$.)

13. (師大微乙 (一)107#1) Evaluate the integrals.

(a) $\int \frac{2x + 1}{\sqrt{4 - x}} dx$

(b) $\int \frac{1}{\sqrt{x}(1 + \sqrt{x})} dx$

(c) $\int x e^{2x} dx$

(d) $\int \frac{-x^3 + 5x^2 - 4x + 9}{x^4 - 2x^3 + 3x^2 - 4x + 2} dx$

(e) $\int \frac{x}{x^2 - 2x + 3} dx$

(f) $\int \tan x \cdot \cos(2x) dx$

(g) $\int_0^{\sqrt{3}} \frac{x^2}{(4 - x^2)^{3/2}} dx$

(h) $\int \sqrt{1 + x^{-1}} dx$

14. (師大微乙 (一)107#5) Find the improper integral

$$\int_0^4 \frac{\ln x}{\sqrt{x}} dx.$$

15. (師大微乙 (一)106#1) Evaluate the integrals.

(a) $\int (x+1)5^{(x+1)^2} dx$

(b) $\int \frac{2x^3 - 4x^2 - 15x + 5}{x^2 - 2x - 8} dx$

(c) $\int e^{\sqrt{2x}} dx$

(d) $\int e^x \sqrt{1 - e^{2x}} dx$

(e) $\int \sec^4 x \tan^3 x dx$

(f) $\int \frac{\sin x}{\cos x - \cos^2 x} dx$

(g) $\int \frac{\ln \sqrt{x}}{x} dx$

(h) $\int_0^1 x \ln x dx$

16. (師大微乙 (一)106#5) 給定下列之瑕積分 (improper integral)，請做以下兩小題。

$$\int_1^\infty \left(\frac{c}{x^2} - \frac{1}{3x} \right) dx$$

(a) 找出使得上述瑕積分收斂 (converge) 的 c 值。

(b) 求 (a) 答案 c 值對應之瑕積分值。

17. (師大微乙 (一)105#1) Evaluate the integrals.

(a) $\int_1^4 (3 - |x - 3|) dx$

(b) $\int \frac{3}{x^2 + 3x - 10} dx$

(c) $\int \sin^2 \theta \tan \theta \sqrt{\cos \theta} d\theta$

(d) $\int \frac{\sec x \tan x}{2 \sec x - 1} dx$

(e) $\int \frac{\ln x}{x^3} dx$

(f) $\int \frac{\sqrt{x^2 - 25}}{x} dx$

(g) $\int \frac{x^3 e^{x^2}}{(x^2 + 1)^2} dx$

(h) $\int_0^2 \frac{1}{\sqrt[3]{x-1}} dx$

18. (師大微乙 (一)104#1) 計算下列積分：

(a) $\int_0^1 (x^3 + 2x) \, dx$

(b) $\int \frac{2x+1}{x^2-7x+12} \, dx$

(c) $\int_0^{\pi/2} \cos^3(x) \, dx$

(d) $\int_1^e \frac{dy}{y(1+(\ln y)^2)}$

(e) $\int \sec^3(z) \, dz$

(f) $\int_1^9 \frac{\log_3 x}{x} \, dx$

(g) $\int \sin \sqrt{2x} \, dx$

(h) $\int \frac{ds}{1+e^s}$

19. (師大微乙 (一)104#5) 判斷下列瑕積分收斂或發散 (請標明使用何種檢定法).

$$\int_1^{\infty} \frac{1}{\sqrt{x^6+1}} \, dx$$

20. (師大微乙 (一)103#1) 底下共有 8 題積分計算題

(a) $\int_1^4 x^2 + 3x + 1 \, dx$

(b) $\int \frac{1}{x^2-5x+6} \, dx$

(c) $\int \sin^3 x \cos^4 x \, dx$

(d) $\int \tan^2 x \sec^3 x \, dx$

(e) $\int \frac{\sqrt{1-x^2}}{x} \, dx$

(f) $\int \ln x \, dx$

(g) $\int \frac{\ln x}{x} \, dx$

(h) $\int \frac{2z}{\sqrt[3]{z^2+1}} \, dz$

21. (師大微乙 (一)103#5)

(a) 驗算 $\int_0^2 x^2 \, dx + \int_0^4 \sqrt{x} \, dx = 8$

(b) 求 $\int_0^{\frac{\pi}{2}} \sin x \, dx + \int_0^1 \sin^{-1} x \, dx$

(d) 判斷下列瑕積分收斂或發散，若收斂請計算其值。

$$\int_0^{\infty} \frac{e^{\tan^{-1} x}}{1+x^2} \, dx$$

22. (師大微乙 (一)102#1) 計算下列積分:

(a) $\int_0^1 (x^2 + 3x + 2) \, dx$

(b) $\int \frac{1}{x^2 + 3x + 2} \, dx$

(c) $\int_0^1 x\sqrt{1-x} \, dx$

(d) $\int_0^{2/3} \sqrt{4-9x^2} \, dx$

(e) $\int \tan^3 x \sec^4 x \, dx$

(f) $\int \frac{1 + \sin x}{\cos^2 x} \, dx$

(g) $\int x^2 e^{-x} \, dx$

(h) $\int \sin \sqrt{x} \, dx$

23. (師大微乙 (一)102#5) 判斷下列瑕積分收斂或發散，若收斂，計算其值。

(a) $\int_0^{\infty} \frac{1}{x^2 + 1} \, dx$

(b) $\int_0^{\infty} \frac{1}{\sqrt{x} + 1} \, dx.$

► 洛必達法則與微積分基本定理

24. (師大微乙 (一)114#4) Find the limit if it exists. (求極限若存在)

(a) $\lim_{t \rightarrow (-2)^+} \int_t^2 \frac{5t + 10}{x^2 - x - 6} \, dx$

(b) $\lim_{x \rightarrow 0} \left(\csc^2(x) - \frac{1}{x^2} \right)$

25. (師大微乙 (一)113 暑 #1) Apply the L'Hôpital rule to evaluate the following limits.

(a) $\lim_{x \rightarrow 0^+} x^2 \ln x$

(b) $\lim_{k \rightarrow \infty} \left(1 + \frac{\pi}{k} \right)^k.$

26. (師大微乙 (一)113 暑 #5) Let f be continuous on its domain. Suppose that

$$\int_0^{x^3} f(3t^2 - 6t) dt = \ln x$$

Find the value of $f(2025)$. (★ Hint: $2025 = 27 \times 75$)

27. (師大微乙 (一)113#1) Find the limit if it exists.

(a) $\lim_{x \rightarrow 0} \frac{x^2}{\ln \cos 2x}$

(b) $\lim_{x \rightarrow 0^+} (e^{2x} + 2x)^{\frac{2}{x}}$

(c) $\lim_{x \rightarrow 0^+} \frac{\int_{\sqrt{x}}^0 \sin t^2 dt}{\sqrt{x^3}}$

28. (師大微乙 (一)112#4) Let $F(x) = \int_{x^2}^0 \arctan(t) dt$ for all $x \in \mathbb{R}$.

(a) Find the open intervals on which F is increasing or decreasing.

(b) Evaluate $\lim_{x \rightarrow -\infty} \frac{F(x)}{x^2}$.

29. (師大微乙 (一)110#2) 使用羅畢達法則計算以下的極限. (Apply L'Hopital's rule to evaluate the following limits.)

(a) $\lim_{x \rightarrow 0^+} x \ln x$

(b) $\lim_{x \rightarrow 1} \frac{\cos(\pi x - \frac{\pi}{2})}{x - 1}$

30. (師大微乙 (一)109#2) Evaluate the limit.

(a) $\lim_{x \rightarrow 0^+} \frac{e^x - 1 - x}{x^2}$

(b) $\lim_{x \rightarrow 0^+} (\sin(2x))^{3x}$

31. (師大微乙 (一)107#2) Evaluate the limit

$$\lim_{x \rightarrow 1^+} (\ln x)^{x-1}.$$

32. (師大微乙 (一)106#4) 請用羅必達法則 (L' Hôpital Rule) 求極限 $\lim_{x \rightarrow 5} \frac{\sqrt{25 - x^2}}{x - 5}$.

33. (師大微乙 (一)105#2) 已知 $\int_0^{x^5} f(2t^2 - t + 1) dt = \ln x$, 求 $f(2017)$ 的值.

34. (師大微乙 (一)105#5) 求極限 $\lim_{x \rightarrow 1} \frac{\frac{1}{4} + \int_1^x t \ln t dt}{x \ln x - x + 1}$ 的值.

35. (師大微乙 (一)104#4) 已知 $\int_{x^2}^2 f(t^3) dt = \frac{1}{2} \sin(\frac{\pi}{2}x)$, 求 $f(1)$.

36. (師大微乙 (一)103#5) 已知 $\int_0^{x^2} f(t) dt = \cos \pi x$, 求 $f(2) = ?$

37. (師大微乙 (一)102#6) 令函數 $f(x) = \int_0^{x^2} \cos t^3 + 1 dt$, 求 $f'(x)$.

► 黎曼和與積分式分析

38. (師大微乙 (一)113 暑 #6) Let the function $f(x) = \frac{1}{1+x}$ be defined on the interval $[0, 2]$. Now, divide (分割) the interval $[0, 2]$ into n equal sub-intervals (n 等份的子區間). For each (每個) $k = 1, 2, \dots, n$, let the k -th sub-interval be

$$\left[\frac{2(k-1)}{n}, \frac{2k}{n} \right]$$

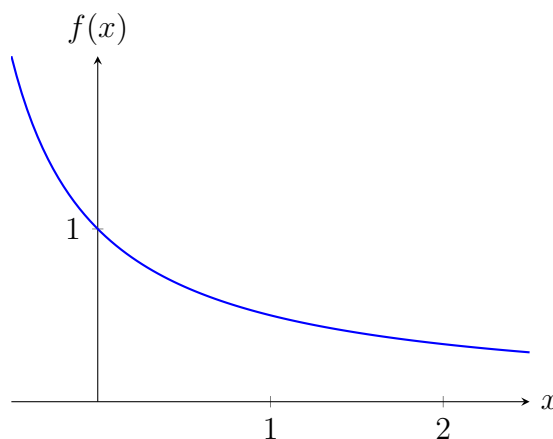
and choose an arbitrary point c_k within the k -th sub-interval (在第 k 個子區間中任意選擇一個點 c_k). Also, define the sequence S_n (數列 S_n) as

$$S_n = \frac{2}{n} [f(c_1) + f(c_2) + \cdots + f(c_n)].$$

(a) (3 pts) Is $S_1 > S_2$ always true? Give your reason!

(b) (3 pts) What is the maximum value of S_4 ?

(c) (4 pts) Find the value of $\lim_{n \rightarrow \infty} S_n$.



(The above figure shows the graph of the function $f(x) = \frac{1}{1+x}$)

39. (師大微乙 (一)113 暑 #7) Let f be a function such that f is differentiable on $[a, b]$ and f' is continuous on $[a, b]$.

(a) Suppose f is one-to-one on $[a, b]$, prove that

$$\int_a^b f(x) dx + \int_{f(a)}^{f(b)} f^{-1}(x) dx = bf(b) - af(a)$$

(★ Hint: Let $y = f^{-1}(x)$, then $y = f^{-1}(x) \iff x = f(y)$)

(b) Evaluate the integral $\int_{11}^{30} \sqrt[3]{\sqrt{x} - 3} - 2 \, dx$.

(★ Hint: Note that $f(x) = \sqrt[3]{\sqrt{x} - 3} - 2$ is one-to-one on $[11, 30]$)

40. (師大微乙 (一)113#7) Evaluate $\lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \left(\frac{7 \cdot 2^k \cdot k^6}{n^7} - \frac{4 \cdot 2^k \cdot k^3}{n^4} + \frac{2}{n} \right) \right)$.

(Hint: The question is about a limit of some Riemann sum (黎曼和). Find the definite integral that the limit represents and evaluate it.)

41. (師大微乙 (一)111#3) Solve the initial value problem for y as a function of x

$$\sqrt{x^2 - 9} \frac{dy}{dx} = 1, x > 3, \quad y(5) = \ln 3.$$

42. (師大微乙 (一)111#4) Consider the well-known standard logistic function(標準型邏輯斯諦函數)

$$f(x) = \frac{1}{1 + e^{-x}}.$$

A typical(典型的) application is a model(模型) of population growth(人口成長), and the logistic functions are widely(廣泛地) used in probability(機率), statistics(統計), machine learning(機器學習), chemistry(化學), social science(社會科學) and so on.

(a) Find all antiderivatives of $f(x)$.

(b) Show that $f(x)$ is a solution of the first-order non-linear ordinary differential equation(一階非線性常微分方程)

$$\frac{dy}{dx} = y(1 - y)$$

with the initial value condition $y(0) = \frac{1}{2}$.

43. (師大微乙 (一)111#5) Consider the error function(誤差函數)

$$F(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

in areas(領域) of probability(機率) and statistics(統計).

(a) Is $F(x)$ continuous?

(b) Does the graph of $F(x)$ have horizontal tangent lines?

(c) Does the graph of $F(x)$ have inflection points(反曲點)?

(d) (Bonus) Does the graph of $F(x)$ have horizontal asymptotes(水平漸近線)?
(Hint: The direct comparison test below will be useful.)

44. (師大微乙 (一)110#6) (Bonus) 證明下列積分值相等 (Prove the following identity.)

$$\int_0^{\pi/2} \sin^n x \, dx = \int_0^{\pi/2} \cos^n x \, dx$$

其中 n 是正整數. (where n is a positive integer.)

45. (師大微乙 (一)108#2) Evaluate the following limit

$$\lim_{n \rightarrow \infty} \frac{\sqrt{n^2 + 1} + \sqrt{n^2 + 4} + \sqrt{n^2 + 9} + \cdots + \sqrt{n^2 + (n-1)^2}}{n^2}$$

by using an appropriate Riemann sum.

46. (師大微乙 (一)106#6) 證明下列積分值是相等的.

$$\int_0^1 x^a (1-x)^b \, dx = \int_0^1 x^b (1-x)^a \, dx$$

其中 $a, b > 0$.

47. (師大微乙 (一)104#6) 給一積分式，如下：

$$\int_a^b \sqrt{x-x^2} \, dx$$

其中 $a, b \in [0, 1]$. 式之最大值為何？請說明原因.

48. (師大微乙 (一)102#8) 求 $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \sqrt{1 + \frac{k}{n}}$ (將里曼和 (Riemann sum) 之極限表成定積分，再求出定積分的值.)

► 計算面積或體積問題

49. (師大微乙 (一)114#1) Let D be the plane region enclosed by the graph of curves $y = \sqrt{9-x^2}$, $x = -1$, $y = 0$ and $y = x$. (設 D 是平面上由這些曲線的圖形圍繞而成的區域)

(a) Find the area of D .

(b) Find the volume of the solid generated by revolving D about x -axis. (求 D 繞 x 軸旋轉所生成之物體體積)

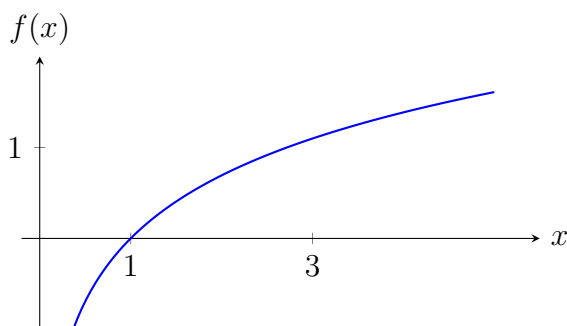
50. (師大微乙 (一)113#3) Find the volume of solid generated by revolving the region bounded by the semicircle $y = \sqrt{4-x^2}$ and the x -axis about the line $y = -1$.

51. (師大微乙 (一)113#4) Let D be the plane region enclosed by $y = -\sqrt{x}$, $x = y$, $y = -2$.

(a) Sketch the region D .

- (b) Find the volume of the solid generated by revolving the region D about the x -axis.
52. (師大微乙 (一)113#5) Find the area of the surface generated by revolving the curve $x = 2\sqrt{8-y}$, $0 \leq y \leq 5$, about y -axis.
53. (師大微乙 (一)112#3) Let $\mathcal{R}$ be the region bounded by the curves $x^3 - x + y = 0$, $y = 0$, $x = 0$ and $1 - x = 0$. Use the Shell Method to find the volume of the solid generated by revolving $\mathcal{R}$ about the y -axis.
54. (師大微乙 (一)112#6) Let Ω be the region bounded by the curves $y = x^2$ and $y = x + 2$.
- (a) Find the area of the plane region Ω .
- (b) Use the Washer Method to find the volume of the solid generated by revolving Ω about the x -axis.
55. (師大微乙 (一)111#2) Find the volume of the solid generated by revolving the region bounded by the curve $y = x^2$ and the line $y = x + 2$ about the line $x = 2$.
56. (師大微乙 (一)110#3) 計算由 $y = 2 - x^2$, $x = 0$, $x = 1$ 和 $y = 0$ 所圍的區域，繞 y -軸所得旋轉體的體積。(Find the volume of the solid formed by revolving the region bounded by the graphs of $y = 2 - x^2$, $x = 0$, $x = 1$ and $y = 0$ about y -axis.)
57. (師大微乙 (一)109#4) Let D be the region bounded by the graphs of $y = x^2 + 1$, $x = 0$, $y = x$ and $x = 1$.
- (a) Find the volume of the solid generated by revolving the region D about the y -axis.
- (b) Find the volume of the solid generated by revolving the region D about the line $y = -1$.
58. (師大微乙 (一)108#5) Let Ω be the plane region bounded by the graphs of $x + y^2 - 6y + 5 = 0$ and $x - y + 1 = 0$.
- (a) Find the area of the region Ω .
- (b) Find the volume of the solid formed by revolving Ω about the x -axis.
- (c) Find the volume of the solid formed by revolving Ω about the y -axis.
59. (師大微乙 (一)107#4) Let D be the plane region bounded by $y = -\sqrt{x+2}$, $y = x$, $y = 0$. Plot the region D and use the shell method to find the volume of the solid generated by revolving the region D about the x -axis.
60. (師大微乙 (一)106#2) 求由 $y = 4x^2$, $x = 0$ 及 $y = 4$ 在第一象限 (first quadrant) 所圍成的封閉區域對 x 軸旋轉所得到的旋轉體體積 (volume).

61. (師大微乙 (一)105#3) 給定由曲線 $y = x\sqrt{x+1}$ 與 x 軸所圍成的區域，試做下列各題：
- (a) 求該封閉區域的面積。
 - (b) 求該區域對 x 軸旋轉所得到的旋轉體體積。
 - (c) 求該區域對 y 軸旋轉所得到的旋轉體體積。
62. (師大微乙 (一)104#2) 函數 $y = \ln x$ 和 x -軸在區間 $1 \leq x \leq e^3$ 所圍出區域。
- (a) 繞 x -軸旋轉的體積。
 - (b) 繞 y -軸旋轉的體積。
63. (師大微乙 (一)104#7) 函數 $y = \ln x$ 和 x -軸在區間 $1 \leq x \leq e^3$ 所圍出區域，繞 $x = 1$ 旋轉的旋轉體體積。
64. (師大微乙 (一)103#2) 求兩條曲線 $y = 7 - 2x^2$ 和 $y = x^2 - 2$ 所圍出區域的面積。
65. (師大微乙 (一)103#3) 令曲線 $y = \sin x$ 和 x -軸在區間 $0 \leq x \leq \frac{\pi}{2}$ 所圍出的區域，繞 x -軸旋轉所得出的實體為 V ，求 V 的體積。
66. (師大微乙 (一)102#2) 求函數 $y = \ln x$ 與 x -軸在 $1 \leq x \leq 3$ 圍出的區域面積 (如圖)。



67. (師大微乙 (一)102#2) 令曲線 $y = \sin x$ 和 x -軸在區間 $0 \leq x \leq \pi$ 所圍出的區域，繞 y -軸旋轉所得的旋轉體為 V 。求旋轉體 V 的體積。
68. (師大微乙 (一)102#7) 曲線 $y = x^2$ 和 $y = 4$ 圍出的區域繞 $y = 4$ 旋轉。求旋轉體體積。

► 計算弧長或表面積

69. (師大微乙 (一)114#2) Find the area of the surface generated by revolving the curve $y = \sqrt{2x - x^2}$, $1 \leq x \leq 1.5$, about x -axis. (求曲線繞 x 軸旋轉所生成之曲面面積)
70. (師大微乙 (一)112#1) Find the area of the surface generated by revolving the following curve

$$x = 2\sqrt{4 - y}, \quad 0 \leq y \leq \frac{15}{4}$$

about the y -axis.

71. (師大微乙 (一)110#4) 求曲線 $y^2 = \frac{4}{9}(x+1)^3$ 由 $x = 2$ 到 $x = 7$ 的弧長.

72. (師大微乙 (一)109#3) Find the arc length the graph of $y = \frac{1}{2}(e^x + e^{-x})$, $0 \leq x \leq 2$.

73. (師大微乙 (一)107#3) Find the arc length of the graph of the function

$$y = \frac{x^5}{10} + \frac{1}{6x^3}, \quad 1 \leq x \leq 3.$$

74. (師大微乙 (一)106#3) 求曲線 $y = \int_{\pi/4}^x \cot t \, dt$ 在 $\frac{\pi}{4} \leq x \leq \frac{3\pi}{4}$ 範圍內的長度 (arc length).

75. (師大微乙 (一)105#4) 求線段 $y = 1 - \frac{x^2}{4}$, $0 \leq x \leq 2$ 對 y 軸旋轉所得到的旋轉體表面積.

76. (師大微乙 (一)104#3) 線段 $y = 2\sqrt{x}$, $1 \leq x \leq 2$ 對 x -軸旋轉的表面積.

77. (師大微乙 (一)103#4) 求曲線 $y = \int_0^x \tan t \, dt$, $0 \leq x \leq \frac{\pi}{4}$ 的長度.

78. (師大微乙 (一)102#4) 求曲線 $y = \int_0^x \sqrt{(\cos 4t)} \, dt$, $0 \leq x \leq \frac{\pi}{4}$ 的長度.

► 計算平均值

79. (師大微乙 (一)113 暑 #4) Find the average value of $g(x) = |x| - 1$ on the interval $[-1, 3]$.

80. (師大微乙 (一)112#5) Find the average value of $g(x) = \frac{1}{e^x + e^{-x}}$ on the interval $[-\ln 3, 0]$.

81. (師大微乙 (一)108#1) Find the average vale of $f(x) = \frac{1}{x^2 - 4x + 7}$ on the interval $[1, 2]$.